What factors best predict university success in the QS World University Rankings 2025?

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1 Abstract

This article explores predictors of university success in the QS World University Rankings 2025 and encompasses 1,503 universities and 28 variables. Three features of university success were identified: employer reputation, ranking tier, and sustainability performance. Using multiple linear regression, the Academic Reputation Score was identified as the best predictor of the Employer Reputation Score. The reduced model achieved an R^2 of 0.777 with an RMSE of 10.43. Using a decision tree, the structural characteristics of the ranking tiers were classified with weak success. Random forest modelling achieved an accuracy of 0.8714, globally classifying the variables regarding sustainability performance in the top quartile, indicating a high class performance based on average accuracy. Logistic regression analysis to identify key features determining a Top 100 ranking achieved an accuracy of 0.95 and ROC of 0.992.

2 Introduction

University rankings are widely used by students, universities, employers, and policymakers to compare institutional performance. According to Alexa Internet [1], QS World Rankings was the most widely viewed university ranking worldwide in 2020. However, "university success" is not a plain concept. A university may be considered successful because it has strong employer recognition, reaches an elite ranking threshold, or performs well in areas such as sustainability. Therefore, this project investigates university success as a multidimensional outcome rather than relying only on the final ranking position.

Existing analysis of the QS World University Rankings primarily uses critical data evaluation to expose metric biases, and machine learning models like neural networks have also been employed for strategic forecasting. One study generated a neural network model predicting university partners' overall scores with accuracy of 98-99% [2]. Our project then aims to extend on this work using the aforementioned predictive analysis tools to more holistically evaluate different dimensions of university success.

3 Methodology

The analysis was based on the QS World University Rankings 2025 dataset [3]. Before modelling, the dataset was cleaned and variables were selected according to the aim of each predictive task. Where possible, score-based indicators were preferred over rank-based variables, because scores preserve the magnitude of differences between universities, while ranks only represent order. Missing values were handled separately within each analysis depending on the variables required for the model.

The project is divided into four predictive analyses. Each section used a different analysation method, including multiple linear regression, logistic regression, decision tree and random forest regression. These analyses aim to show which types of variables are most useful for predicting different dimensions of university success in the QS 2025 dataset.The project is divided into four predictive analyses. Model performance was evaluated using metrics appropriate to each task. Training, validation, and test splits were used to evaluate performance on unseen data and reduce the risk of overfitting.

4.3 Random Forest - Jawad Uddin - Can institutional characteristics & performance indicators predict sustainability performance?

4.3.1 Aim & Methodology

This section focuses on predicting sustainability performance in the QS World University Rankings, with the question sub-dividing into two categories of predictors:

- **Institutional Characteristics**, e.g. region, size, and primary subject focus.
- **Performance Indicators**, e.g. academic reputation, international faculty, and international research network.

This division has intentionally been chosen to suit the interests of key stakeholders in this analysis, such as policymakers and strategic planners, by distinguishing between fixed structural realities and adaptable performance metrics. This enables stakeholders to independently compare the impact of largely unchangeable institutional characteristics with that of the performance metrics (where targeted investment and policy interventions can change sustainability outcomes).

Random Forest is an ensemble machine learning algorithm which aggregates the predictions of many decision trees, each generated by the random row-column selection process formally known as feature bagging. This makes it less prone to overfitting by essentially limiting the feature space that each tree sees. [13]

4.3.2 Filtering, Splitting, Dependent Variable & Model Selection

The target variable is the Sustainability Score, a numeric score between 0-100 where 100 represents the highest sustainability performance. Data cleaning was applied to filter out rows with missing sustainability scores. The continuous target variable calls for a regression model, but we recognise that being able to predict numeric scores is less generalised. We thus introduce a binarised variable:

$$Y_i = \begin{cases} 1 & \text{if } S_i \geq Q_3 \\ 0 & \text{if } S_i < Q_3 \end{cases} \tag{2}$$

which indicates whether a given university falls into the top quartile of sustainability scores. The top quartile was selected because it identifies a meaningful group of high-performing institutions, while retaining a sufficient number of observations for model training. The Random Forest Classifier analysis is intended to isolate the key 'tipping point' required to break into the top 25% (i.e. beyond just identifying a linear relationship - looking at features that consistently act as strict gatekeepers to this threshold). This empowers policy-shapers with the evidence to say "without X change, we likely won't achieve Y sustainability outcomes."

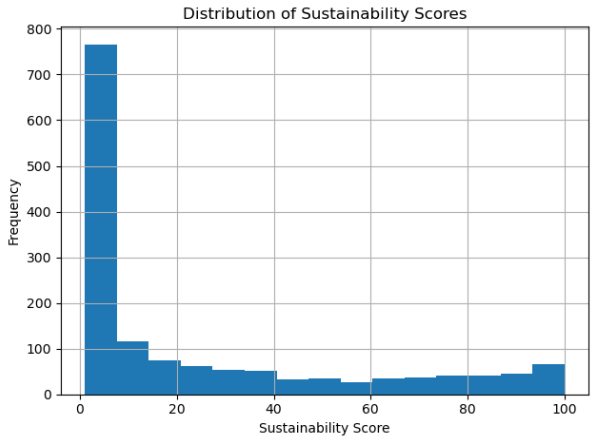


Figure 9: Histogram of Sustainability Scores

An initial look into the distribution of scores reveals a disproportionately large number of values at the lower tail - this imbalance might affect the accuracy of a regressor model and further justifies our choice of selecting the top 25% for classification.

We will focus on the Classifier's results and use the Regressor in a secondary capacity, as a more granular lens to compare and contrast the continuous nuances of the data. There is a 3:1 ratio in negative to positive results, but undersampling the majority class cuts the dataset in half, so we instead will use class weighting to ensure that mistakes on positive results are more costly than those on negative results. Finally, we will randomly split the data into 80% training and 20% test. For validation, we perform hyperparameter tuning using 5-fold cross-validation on the training set (leaving the test set exclusively for evaluation).

4.3.3 Hyperparameter Tuning

We tuned four hyperparameters of the Random Forest Classifier - n_estimators, max_depth, min_samples_leaf, and class_weight. This was conducted using a GridSearchCV to iterate through reasonable initial guess ranges for these variables. These 4 hyperparameters were selected due to their higher projected effect on model accuracy. Other parameters like min_sample_split and min_impurity_decrease were considered but ultimately excluded. These would have introduced unnecessary computational complexity, and the Random Forest, whose ensemble nature

makes it naturally more resistant to overfitting, would see less impact from these.

N Estimators

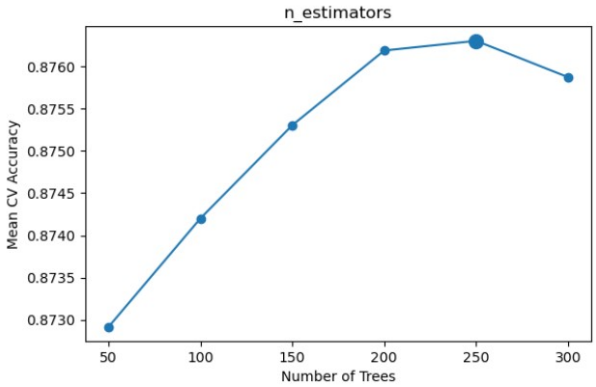


Figure 10: Plot of Model Accuracy against N Estimators

N Estimators represents the number of trees in the model. Typically, the accuracy from training increases with more trees, but with diminishing returns, as shown by the plateau in the above figure, We selected a value of 250 as the best compromise between accuracy and speed of training.

Maximum Depth

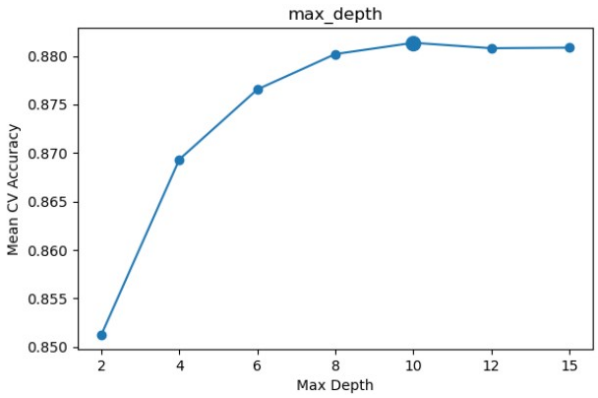


Figure 11: Plot of Model Accuracy against Maximum Depth

This parameter controls, for each tree, the longest allowable path between the root and leaf node. The above graph shows that increasing tree depth also increases the model accuracy (due to being able to capture finer detail with more stages) - however this again reaches a plateau due to the risk of overfitting. We ultimately chose a depth value of 10 to ensure good generalisability.

Minimum Samples per Leaf Node

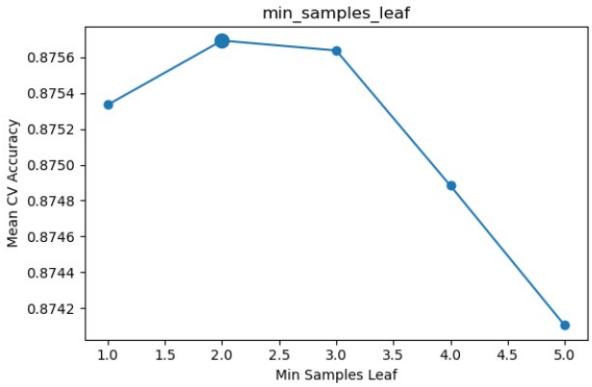


Figure 12: Plot of Model Accuracy against Minimum Samples per Leaf Node

Imposing a minimum number of samples per leaf can prevent splits that enforce overly specific rules. Low values result in heavily complex trees that are more prone to overfitting, whereas increasing the minimum forces trees to generalise more. Interestingly, we observed a decrease in accuracy after the 2-sample mark, which could suggest underfitting considering the depth of trees and total number of samples.

Class Weighting

In our GridSearchCV, we specified two values to compare for this hyperparameter: None, and Balanced. As expected, balanced class weighting far outperformed the alternative, achieving an averaged accuracy of 0.8714.

4.3.4 Results

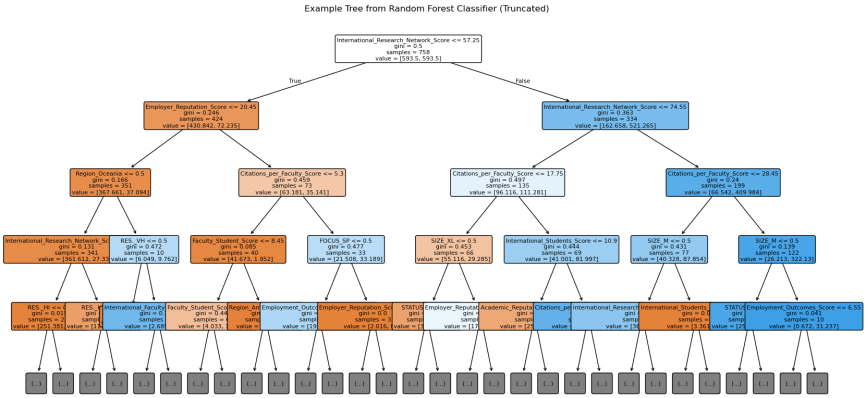


Figure 13: Example Tree from Random Forest Classifier (truncated to depth 4)

Sustainability Category	Precision	Recall	f1-Score
Non-Leader (bottom 75%)	0.95	0.93	0.94
Leader (top 25%)	0.81	0.85	0.83

Table 7: Testing Set Results from Random Forest Classifier

Metric	Training	Testing	Difference
Accuracy	0.9764	0.9125	0.0639
AUC	0.9976	0.9656	0.0320

Table 8: Results from Random Forest Classifier

To assess potential overfitting, model performnace was compared for the training and testing set. The higher accuracy and AUC in training, with a small decline moving towards testing, suggests a moderate degree of overfitting, but the decline in performance is relatively small and suggests that the model still generalises well to unseen university data. A final AUC of 0.9656 and accuracy above 90% shows that the classifier still captures meaningful relationships.

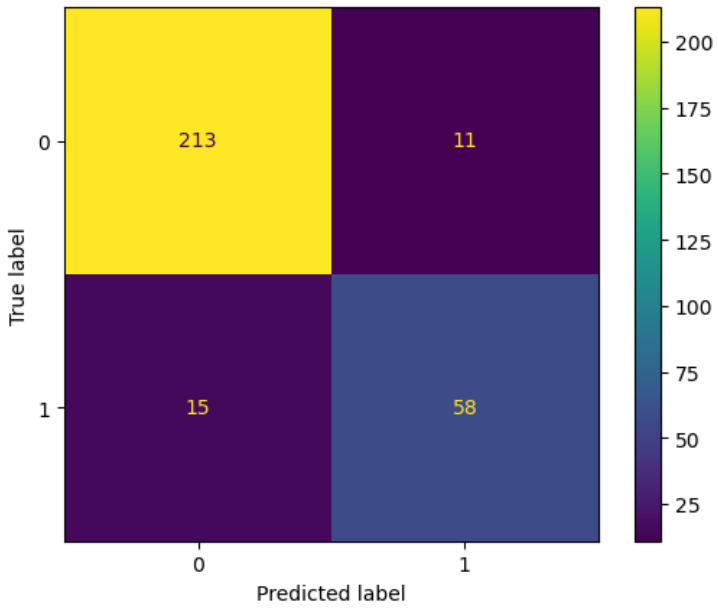


Figure 14: Confusion Matrix

The confusion matrix shows that a large proportion of the testing set fell into the true positive-negative diagonal, with 11 true non-leader universities and 15 true leader universities being incorrectly assigned.

4.3.5 Feature Importance & Discussion

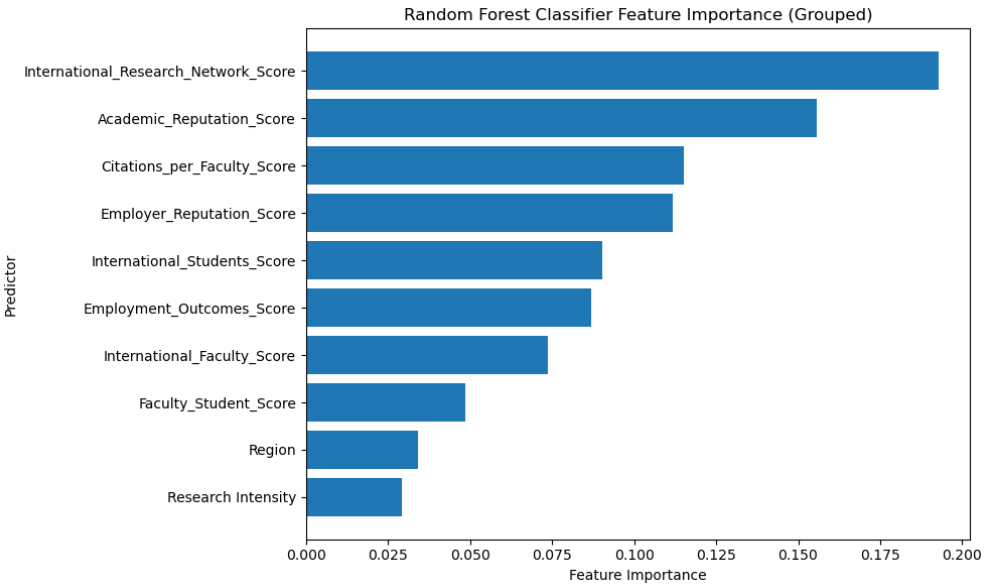


Figure 15: Classifier Feature Importance Plot for 10 most important features

Random Forest can provide a measure of feature importance due to its ensemble nature. Thus, we plotted this for the classifier, finding that the most significant features were International Research Network (19.27%), Academic Reputation (15.57%), and Citations per Faculty (11.51%). Interestingly, all performance indicators placed higher than the institutional characteristics, with the most significant institutional characteristic being region (3.42%). This disparity gives evidence for the adaptable performance indicators having a much higher bearing on sustainability performance than fixed instutional characteristics, by almost one order of magnitude.

The final model had a precision of 0.81, recall of 0.85, and accuracy of 91.25%. For the purposes of university policymakers and strategic planners, this indicates a good model performance, and based on the identified order of feature importance, provides fairly reliable insight on the key markers required to break into the top quartile for sustainability performance in the QS World Rankings.

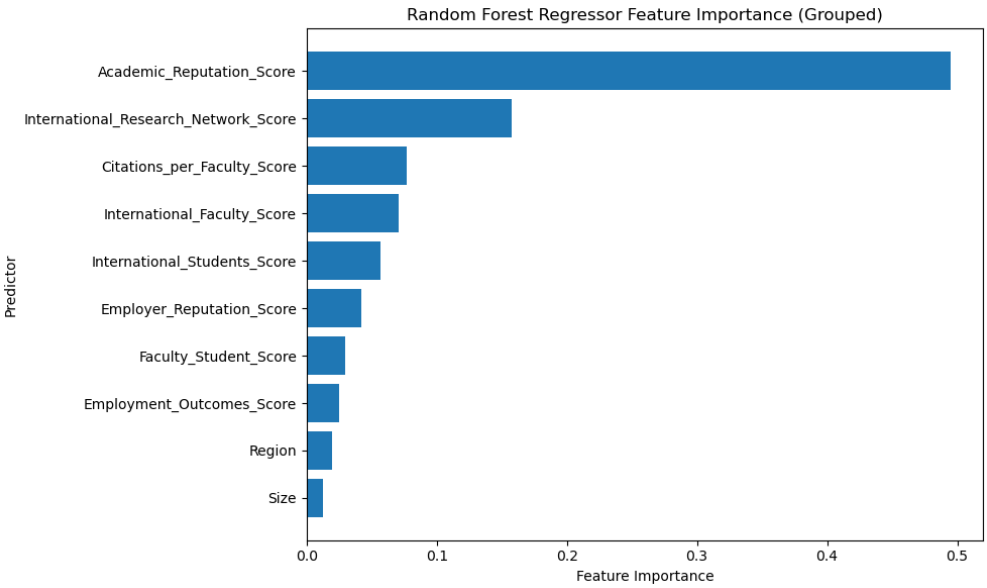


Figure 16: Regressor Feature Importance Plot for 10 most important features

To further validate the feature importance results, a Random Forest Regressor model was subordinately trained and hyperparameter-tuned in a similar fashion, and a similar plot generated. Performance indicators still dominate, though it is interesting to note that the Academic Reputation score here significantly outweighs other indicators, which may be influential for subsequent strategic decisions.

5 Conclusion

In each of the four predictive models, performance-based measures turn out to be the main predictors of university success in the QS World University Rankings 2025 in comparison with institutional attributes. However, despite the fact that various machine learning techniques (e.g., logistic regression, random forest and regression analysis-based models) were used, all models produced the same significant result – the factors of academic reputation and research contribute to institutional success the most.

Such high performance of models may be justified by their high suitability to the problem. For instance, the logistic regression model demonstrated a very high level of classification accuracy (Accuracy = 0.95, ROC-AUC = 0.992, Recall = 0.95), and the forward feature selection produced a compact model, in which Academic Reputation was characterised by the highest coefficient (3.14). Moreover, the random forest model exhibited a good fit for the task (Accuracy = 0.91, AUC = 0.97).

Therefore, agreement of models improves the reliability of results, suggesting that QS success depends on performance more than on institutional characteristics. The results hold relevance for the university policmakers and strategy offices because they are consistent with prior datasets, which have more heavily weighted performance indicators, especially Academic Reputation.

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Appendix

Full code: https://github.com/Jawad12256/QS-World-University-Rankings-2025-Analysis.git